

Deepak Mallampati

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Kent, OH -- open to SF relocation; F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer building healthcare NLP and agentic LLM systems for clinical and revenue-cycle-adjacent workflows. Production track record: ~**35%** retrieval precision gain on clinical RAG, >**30%** hallucination reduction on agentic LLM workflows, **15-20%** recall gains on high-risk patient categories, ~**30%** fewer post-deployment defects. Built **Agentic Healthcare Claims** pipeline -- multi-agent CPT/ICD validation with schema-validated JSON contracts between agents to prevent cascading hallucinations and audit-ready reasoning traces, directly analogous to medical-billing-agent patterns. Hands-on with transformer-based NLP (pre-processing, embeddings, HuggingFace), supervised fine-tuning (Stable Diffusion + LoRA), LLM evaluation pipelines, and FastAPI/Docker production deployment in HIPAA-conscious environments.

CORE COMPETENCIES

Transformer-Based NLP . HuggingFace Transformers & Diffusers . Supervised Fine-Tuning . LLM Evaluation Pipelines . Agentic LLM Systems . Production RAG & Grounding . Healthcare NLP & CPT/ICD Validation . HIPAA-Conscious Governance . Python / PyTorch / FastAPI . Docker & Production Discipline . End-to-End Ownership

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | Healthcare Technology, AI Consulting

- Built **transformer-based NLP RAG systems** for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking with HuggingFace transformer embeddings; improved retrieval precision ~**35%** and reduced irrelevant retrieval >**30%**; grounded-response alignment exceeded **90%** in evaluation harness.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent stability ~**25%** and reduced hallucination rate >**30%**.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for patient no-show prediction, care engagement scoring, and support prioritization; improved recall **15-20%** on high-risk categories while holding precision >**90%** via class weighting, stratified sampling, and threshold calibration.
- Packaged LLM/ML inference as **FastAPI/Flask** RESTful services containerized with Docker; structured logging + load simulation cut post-deployment defects ~**30%**.
- Maintained HIPAA-conscious governance: de-identification, data lineage docs, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India | Enterprise ERP, Compliance-Sensitive

- Optimized T-SQL stored procedures powering the Synthesis Order-to-Cash platform; reduced query execution ~**20%** and data retrieval latency ~**25%**.
- Built CI/CD pipelines with Jenkins for schema updates and stored procedure deployments; reduced deployment errors >**30%** and improved release cycle ~**35-40%**.
- Supported RBAC and audit logging for financial modules in compliance-sensitive oil & gas environment.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Nonprofit Tech, Applied NLP

- Built **transformer-based video summarization** system across 5,000+ recorded sessions; transcript preprocessing, hierarchical abstractive summarization for long-context, timestamp-aligned clip extraction; reduced manual review time **60-70%**; ~**85% alignment** with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%**; raised contextual answer accuracy >**85%**.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent NLP pipeline (Intake Normalization -> Validation -> Consistency Analysis -> Duplicate Detection -> Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations -- direct analog to medical-billing-agent patterns. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity for duplicate-claim detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, transformer embeddings, vector search, FastAPI, Docker, healthcare ontology (CPT/ICD)

Agentic Pixel -- Character Synthesis & Animation Engine

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis. **Stable Diffusion supervised fine-tuning + LoRA** for identity consistency, ControlNet + OpenPose/MediaPipe for pose conditioning, and an **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks.

PyTorch, HuggingFace Diffusers, Stable Diffusion, LoRA fine-tuning, FastAPI, Docker

Dream Decoder -- AI Multimodal Platform

Full-stack FastAPI + React/TS app with coordinated multimodal APIs. Introduced **intermediate structured prompt transformation layers** -- improved contextual alignment **~30%** and first-pass image success rate **~25-30%** over naive direct-prompt orchestration.

FastAPI, React, TypeScript, structured prompt layers, multimodal APIs

Manga Lens shipped -- Chrome Web Store

Solo-built Chrome extension integrating **4 AI vision providers** (Claude, GPT-4o, GPT-4.1 Nano, Ollama). Multi-section capture pipeline, 7-day cache, 29 site selector configs, scoped permissions, Chrome review-ready.

TypeScript, Manifest V3, content scripts, multi-provider abstraction

EDUCATION

Master's Degree -- Kent State University, Ohio, USA

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL, SQLite), C++ (Arduino)

NLP & Transformers: HuggingFace Transformers, HuggingFace Diffusers, transformer-based embeddings, pre-processing, hierarchical abstractive summarization, supervised fine-tuning (Stable Diffusion + LoRA), LLM evaluation pipelines, prompt engineering, structured outputs, guardrails, grounding

ML Libraries: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, Pandas, NumPy

Healthcare AI: EHR data, HIPAA-conscious governance, de-identification, claims processing (CPT/ICD), audit trails, system-limitation docs

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging, cloud-ready deployment

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required